

MEMORANDUM

To: Newcastle Planning Board (*working draft for Ben Frey*)
From: Ben Frey (*Article 3 / Core Zoning Code amendment drafter*)
Date: June 9, 2026
Re: Reconciling two parallel “Thoroughfare” efforts — our form-based **Article 3 (Thoroughfares)** vs. the Fall-2026 redline’s engineering-style **“Article X: Thoroughfare Standards”**
Status: **Discussion draft.** Compares two in-progress drafts. No code language is changed here.

1. PURPOSE

Two separate drafting efforts are independently writing thoroughfare standards for the CZC for the same amendment cycle:

- **Ours** — *Article 3, “Thoroughfares”* (this project): a **form-based** street/road system (Types **S1-S5** Streets, **R1-R5** Roads), calibrated to the zoning **Districts**, with a binding classification of every existing segment.
- **Theirs** — the Fall-2026 redline’s *“Article X: Thoroughfare Standards”*: a **conventional engineering** road code (13 functional types in two design-standard tables) plus new **definitions** of Thoroughfare/Street/Road/Driveway/Right-of-Way.

Both **replace the same repealed instrument — the Road, Driveway & Entrance Ordinance (RDEO)** — and, as shown below, both lift engineering text nearly verbatim from it. This memo maps where they agree, where they conflict, what each uniquely contributes, what each is missing, and which approach better “fits” the CZC.

Bottom line up front: the two cannot both be adopted — they define the same core terms in contradictory ways and would produce an internally inconsistent code. Our form-based article is the better structural **fit** for a form-based code and is, in most respects, a **superset**; their Article X contributes a short list of engineering specifics worth harvesting. The recommended path is **merge, not choose**: keep our framework, graft in their best engineering details, reconcile the definitions deliberately.

2. THE TWO ARTIFACTS AT A GLANCE

	Our Article 3 (Thoroughfares)	Their “Article X: Thoroughfare Standards”
Lineage	SmartCode / form-based (transect)	Subdivision-road engineering (RDEO modernized)
Organizing axis	Zoning District (place / transect)	Function & size (and use restriction)
Types	10 , place-named (Main/Village/Neighborhood/Lane/Alley; Connector/Rural/Rural-Lane/Highway-Commercial/Rural-Highway)	13 , function-named (Track/Small/Advisory/Local/Rural/Industrial Rd; Advisory/Local/One-Way/Connector/Industrial St + Alley; “Main St” struck)
Applies to	All thoroughfares — existing (via a binding map) and new	New / substantially reconstructed only
Form of standards	Illustrated cross-section “plates” + 14-row standards table per Type	Two dimensional tables (X.1 Roads, X.2 Streets)
Integration	Wired into Districts, frontages, setbacks, shopfronts, Comp Plan	Stand-alone; placeholder number; driveways punted to “Article... Site Standards”
Maturity	Complete, rendered, classified (215 segments)	Draft; several table cells blank; “Article X” un-numbered

3. CONCEPTUAL / PHILOSOPHICAL COMPARISON

- **Ours treats the street as urban form** — part of the public realm that shapes (and is shaped by) the abutting District. The S/R Types are a **transect**: S1 Main Street in the Town Center (D6) grading down to R5 Rural Highway on the rural Route 1 corridor. Each Type is a **cross-section** (sidewalk · planting · parking · travel · shoulder · verge) with a design speed, a max grade, a max block length, and build-to expectations. The street is **classified and mapped** for every existing segment, and assigned by District for new ones.
- **Theirs treats the road as infrastructure to be engineered** — how to build a road that drains, lasts, and carries its intended traffic. Its types are **functional/size classes** with paving material, shoulder, underdrainage, culverts, ditches, sight distance, hammerheads, and **per-type use/volume/vehicle restrictions** (e.g., “residential only, ≤8 dwelling units, no trucks”). It applies at **construction/subdivision review**, not as a town-wide classification.

These are **two different jobs**: ours answers “*what should this thoroughfare be, and where?*”; theirs answers “*how do you build one that works?*” That is exactly why a merge is natural — and why **both descend from the RDEO**, so their engineering cores substantially overlap (see §6).

4. TERMINOLOGY & DEFINITIONS — THE HARD CONFLICT

Both drafts start from the **same three original CZC definitions** (Road, Driveway, Right-of-Way) and **diverge in incompatible directions**. This is the single biggest reconciliation problem: adopting both would put two contradictory definitions of the same words in one code.

Term	Original CZC	Ours	Theirs	Verdict
Thoroughfare	(none)	Umbrella for any Street, Road, or Driveway ; all 10 Types + Driveway	“A type of right-of-way , with its improvements, that accommodates passage and provides access to adjacent properties”	Conflict. Ours includes Driveway; theirs is ROW-based and excludes driveways.
Street	(none)	A thoroughfare in the S1-S5 family ; pedestrian comfort primary	“A category of thoroughfares for ‘in-town’ settings , greater/varied volume, more improvements, higher standards”	Compatible in spirit (both = urban), different mechanism (family vs. setting).
Road	Physical “route, track, or way” of soil/gravel/asphalt	(kept the physical def — see §13 note)	“A category of thoroughfares for ‘rural’ settings , minimal improvements, use restrictions”	Conflict — and theirs is arguably more internally consistent than ours here.
Driveway	Access from a Public/Private Road to a lot; not built to road specs	(kept original)	“A route or track... not within a right-of-way ”	Conflict.
Right-of-Way	“Total width of land reserved/dedicated as a road, alley, ped/bike way...”	(kept original)	“A type of easement with defined boundaries...” + Public / Private ROW	Conflict — theirs is the cleaner legal framing (ROW as easement).

Naming collisions in the Types, too: both use “**Connector**” (ours = R1 Connector Road, rural; theirs = Connector Street, urban, 42 ft) and both use “**Alley**” and “**Main St**” (ours central; theirs struck). Any merge must de-collide these.

5. THE TYPOLOGIES, SIDE BY SIDE

Best-fit mapping by right-of-way and function (ROW = minimum width):

Ours	ROW	≈ Theirs	ROW	Notes
S5 Alley	16-20	Alley	20	Closest 1:1 match.
S4 Lane	30-40	Local St / Advisory St	33	Narrow low-speed street.
S3 Neighborhood St	40-46	Local St	33	Ours wider (planting + walks).
S2 Village St	40-54	Connector St	42	Mixed-use village street.
S1 Main St	66-80	Main St (struck)	(50)	Theirs deleted Main St → gap; ours much wider.
R3 Rural Lane	33-50	Track Rd / Small Rd	30 / 49	Both honor the historic “two-rod” (33 ft) way.
R2 Rural Rd	40-50	Rural Rd	49	Strong match.
R1 Connector Rd	40-50	Local Rd	49	Theirs adds a sidewalk + parking.
R4 Highway Commercial	MaineDOT (~56)	— none —	—	Theirs has no state-highway type.
R5 Rural Highway	MaineDOT (~68)	— none —	—	Route 1 corridor unaddressed in theirs.
(none)	—	Local One-Way; Advisory (shoulder); Industrial Rd/St	33 / 30	Geometries ours lacks.

Naming philosophy: ours is **place/transect-based** (evokes where you are); theirs is **function/volume-based** (evokes what it does). Both nod to Maine’s historic **rod** survey unit — theirs literally tabulates street ROW in rods (2.00 / 2.55 / 3.03), ours references “two-rod ways” in R3.

6. DESIGN STANDARDS — AND THE SHARED RDEO CORE

The engineering cores are nearly identical because both copy the RDEO. Side-by-side, our §10 (Water Management) and their Article X water section are almost word-for-word:

	Ours (§10)	Theirs (Art. X)
Culvert extension	“...beyond traveled way + shoulders by 18 inches both sides”	<i>identical</i>
Culvert diameter	“minimum 15 inches ”	“minimum 15 inches ”
Culvert placement	“at stream bed level over a prepared base”	“between ¼ and ⅓ of diameter below the stream bed” ← <i>more specific</i>
Fish passage	“unobstructed fish passage either direction”	<i>identical</i>
Ditch flow lines	“ 2 ft below the sub-grade breakpoint”	<i>identical</i>
Ditch seeding	“seeded with erosion control landscaping”	<i>identical</i>

Both also require **engineer-stamped plans** and **MaineDOT Standard Specifications** for construction. So the “engineering vs. form-based” contrast is real at the **framework** level but **overstated at the detail level** — we already carry most of the RDEO engineering in our §10 (Water Management) and §11 (Construction).

Where the dimensional philosophies differ: - **Right-of-way:** ours is **wider / public-realm** (S1 66-80, S3 40-46), theirs **leaner / utilitarian** (streets 33-50, roads 30-49). Ours reserves room for trees + walks both sides; theirs minimizes the take. - **Travel lanes:** both **narrow for calming** (8-12 ft) — genuine agreement. - **What ours tabulates that theirs doesn’t:** design speed, max grade, max block length, curb-return radius, street-tree

requirement by context. – **What theirs tabulates that ours doesn't:** paving **material by type** (gravel/macadam), **underdrainage by type**, and **shoulder width by type**.

7. COVERAGE MATRIX (✓ HAS · ○ PARTIAL · ✗ LACKS)

Capability	Ours	Theirs
Form-based cross-sections / illustrations	✓	✗
Types calibrated to Districts (transect)	✓	✗
Binding classification of existing segments (inventory + Type Map)	✓	✗
Ownership Categories (Town Way / Easement / State Highway / Private), decoupled from Type	✓	✗
State-highway / MaineDOT corridor type + coordination (§12)	✓	✗
Driveways	✓ (§7, by District)	✗ (punted to Site Standards)
Entrances, Intersections, sight-distance	✓ (§8, §9)	○
Water management (culverts/ditches)	✓ (§10)	✓
Construction specs / MaineDOT / engineer-stamped	✓ (§11)	✓
Design speed · max grade · block length · curb radius	✓	✗
Frontage / setback / shopfront integration	✓	✗
Comp-Plan linkage	✓	✗
Per-type use / volume / vehicle restrictions	✗	✓
Paving material & underdrainage by type	○ (defers to tables)	✓
Road-siting spacing (≥30 ft from un-served abutter; ≥50 ft between non-intersecting)	✗	✓
Dead-end hammerhead turnaround (≤150 ft)	○ (mentions dead-ends, no turnaround geometry)	✓
One-way / advisory-shoulder geometries	✗	✓
Easement-recording requirement	○	✓

8. WHERE THE TWO GENUINELY MATCH

1. **Street = urban, Road = rural** as the top-level split.
2. **Typed systems** — named types with per-type dimensional tables (not one-size-fits-all).
3. **Narrow travel lanes** (8–12 ft) for traffic calming.
4. An **Alley** at ~20 ft, and respect for the historic **rod/two-rod** ROW.
5. **Engineer-stamped plans + MaineDOT construction specs** for new roads.
6. **Near-identical culvert/ditch water-management** language (shared RDEO source).

7. **“Thoroughfare”** as the umbrella term, and both **replace the repealed RDEO**.

9. WHERE THEY FUNDAMENTALLY DIFFER

1. **Philosophy:** form/transect (ours) vs. engineering/function (theirs).
2. **Organizing axis:** District/place vs. size/use-class.
3. **Scope:** all thoroughfares incl. existing, via a binding map (ours) vs. new construction only (theirs).
4. **Definitions** of Thoroughfare/Road/Driveway/ROW (direct conflicts — §4).
5. **ROW philosophy:** wider public realm vs. leaner utilitarian.
6. **Use control:** by **District** (ours) vs. by **Type restriction** (theirs).
7. **Driveways:** integrated (ours) vs. separated (theirs).

10. UNIQUE TO EACH

Unique to ours: the §5 **binding classification + Type Map** (every existing segment typed, backed by the GIS pipeline); **District calibration**; **Ownership Categories** (§4); **R4/R5 + §12 MaineDOT** corridor handling; **cross-section illustrations**; design speed / grade / block length; driveway calibration, entrances, intersections; Comp-Plan and form-code integration.

Unique to theirs: per-type **use/volume/vehicle restrictions**; **paving-material & underdrainage by type**; the more exact **culvert placement** ($\frac{1}{4}$ – $\frac{1}{3}$ below streambed); **road-siting spacing** rules; **hammerhead** turnaround; **one-way / advisory** geometries; the **ROW-as-easement** definitional framing (Public/Private ROW).

11. GAPS / WHAT’S MISSING

Missing from ours (harvest from theirs): an explicit **dead-end turnaround/hammerhead** standard; **road-to-road and road-to-lot-line siting spacing**; **paving material & underdrainage by Type**; optionally, **use/volume restrictions** for the lowest rural Types; the cleaner **ROW-as-easement** definition.

Missing from theirs (ours supplies): any **classification of existing roads** / map / inventory; **District guidance** on which type goes where; a **state-highway/MaineDOT** type (Route 1 is unaddressed; **“Main St”** was deleted, leaving no downtown standard); **ownership categories**; design speed / grade / block length / curb radius; **illustrations**; **driveway** standards (punted); frontage/form integration. Several **tables are still blank/placeholder**, and the article is **un-numbered (“Article X”)**.

12. WHICH “FIT” IS BETTER FOR THE CZC

Our form-based Article 3 is the better fit, and by a clear margin — because the CZC is itself a form-based code. Every other part of the code is transect/District-driven: District pages, building types, frontages, setbacks, shopfronts, use-by-District tables. A thoroughfare article that is **District-calibrated and illustrated (ours) speaks the code’s native language** and plugs directly into those systems. The other Article X, however competent as an engineering standard, is **structurally foreign** to a form-based code: it is organized by function and use-restriction rather than by place, has no transect logic, no map, and doesn’t connect to frontages or setbacks — it reads like a subdivision-road appendix bolted on.

But theirs is not redundant. It carries a handful of **engineering specifics and siting rules** that genuinely improve a new-road standard (turnarounds, road spacing, paving/underdrainage by type, the precise culvert placement), plus the **idea of use/volume restrictions** and a **cleaner ROW definition**. And because both descend from the RDEO, grafting is low-friction — the engineering vocabulary already matches.

Recommendation — merge, don’t choose: 1. **Adopt our Article 3 as the framework** (Types, District calibration, §5 classification + map, ownership, MaineDOT corridor, illustrations). 2. **Harvest from Article X** into our §6/§10/§11: the **hammerhead/turnaround** standard, **road-siting spacing**, **paving-material & underdrainage by Type**, and the $\frac{1}{4}$ – $\frac{1}{3}$ **culvert placement**; evaluate **per-Type use/volume restrictions** for R3/Track-class rural ways. 3. **Reconcile the definitions deliberately** — keep our developed Street/Road/Thoroughfare/Type set, but **borrow their ROW-as-easement framing** (Public/Private ROW) and **fix our own “Road” definition** (see §13). 4. **Decollide the names** (“Connector,” “Main,” “Alley”). 5. Treat the **rest of their Fall-2026 package** (permitting/review

overhaul, Project Size Standards, Article 2 changes) as **largely independent** — it can proceed on its own track; only the *thoroughfare* piece collides with us.

Do not let both reach Town Meeting. Two contradictory definitions of Thoroughfare/Road/Driveway/Right-of-Way, two different Type systems, and two different Article-3 numberings would make the adopted code internally inconsistent and litigation-prone. The two efforts must be merged into one article **before** warrant.

13. NOTE FOR OUR OWN CLEANUP (INDEPENDENT OF THE MERGE)

Their redline exposes an inconsistency on **our** side: our Article 9 **“Road”** definition is still the *original physical* one (“a route, track, or way consisting of a bed of exposed mineral soil...”), even though our Article 3 treats **Road as the R1–R5 family** (parallel to how we already rewrote **“Street”** as the S-family). The other planner *did* update “Road” to a category definition. We should align our “Road” definition with our family concept for internal consistency — a small, safe edit worth folding into a future version.

This memo compares two draft documents for coordination purposes; it is not legal advice.